MARCH 19, 2026

Matt | Varghese Mathew

Software Systems Architect

Software Systems Architect with 20 years of experience designing large-scale interoperable health platforms. Delivered Epic's HL7 FHIR-based CMS 2027 regulatory implementation, standardizing prior authorization workflows for hundreds of customers. Recognized for transforming fragmented integrations into unified, scalable platforms through deep domain expertise and rigorous system design.

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Madison, WI

linkedin.com/in/matt-varghese

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WORK EXPERIENCE

Software Systems Architect, Health Plans

Epic - Verona, WI — *MARCH 2021 - PRESENT*

Architect for Epic's HL7 FHIR-based CMS 2027 regulatory platform, defining how health plans implement real-time prior authorization workflows across CRD, DTR, and PAS standards. Systems adopted by hundreds of Epic customers, transforming authorization turnaround from days to near real-time.

- **Led architecture and delivery** of Epic's payer interoperability platform for CMS 2027, collaborating with HL7 Da Vinci workgroups and external vendors to refine and operationalize evolving standards.
- **Established a unified "source of truth" for prior authorization requirements,** eliminating fragmentation across 12+ product modules and reducing new development lead time by ~80%.
- **Architected standardized data models and execution frameworks** to replace inconsistent one-off implementations, significantly improving maintainability and cross-team scalability.
- **Modernized Medical Necessity guideline integrations (MCG, InterQual)** using SMART on FHIR, enabling real-time clinical decision workflows and reducing integration defects by ~60%.
- **Built internal simulation and testing frameworks** to emulate third-party systems, enabling end-to-end validation of FHIR workflows, proactively eliminating defects, and accelerating partner onboarding.

Senior Software Systems Developer, Clinical Informatics

Epic - Verona, WI — *AUG. 2010 - MARCH 2021*

Lead developer and architect for mission-critical clinical decision support integrations and mobile infrastructure. Directed adoption of modern FHIR-based clinical decision support standards and pioneered Epic's first high-availability cloud service on Azure.

- **Led architecture for Epic's mobile push notification broker service** routing millions of weekly notifications via APNS and GCM. Designed for high availability as one of Epic's first Azure-native production services, reducing operational costs by over 90%.

PROJECTS

FHIR-based AI Assistant

2025 Q2

Demonstrates seamless integration of LLM APIs with HL7 FHIR for agentic use-cases.

LLM chat web app to answer provider and patient questions using real EHR data.

Can be leveraged for AI driven forms, clinical decision support, care delivery, etc.

JSON Parser for M/Caché

2020

Spotlights long-term vision in creating powerful generic reusable tooling.

RFC-compliant JSON Serializer and Deserializer for ANSI-M/Caché.

Nearly half of all Epic's database AI integrations leverage this library.

Web Based SSH Client

2025 Q3

Out-of-the-box solutioning for real world cost reduction avoiding remote-app licenses for moderated shell / text access to servers.

Proof of concept fully browser-based SSH client for unix servers, Caché etc.

github.com/mattvarghese/WebSSHClient

HTTP Auto-responder

2024 Q4

Development tooling for Web APIs, keeping traffic within the enterprise or local network.

REST-ful endpoint allowing auto-responding and logging for common HTTP verbs.

github.com/mattvarghese/HttpResponder

MIPS Pipeline Simulator

2005, 2025

Highlights expertise in low-level systems, architecture, parallelism, and performance.

5-Stage pipeline simulation of a MIPS microprocessor, with cache simulation, assembler, and terminal app developed with POSIX APIs, C++, Flex, and Bison in 2005.

In 2025, developed a Small-C compiler above the assembler with AI assistance.

github.com/mattvarghese/coconut

- **Centralized 3rd-party web application integration frameworks** across the enterprise to improve security and maintainability. Developed a suite of validation tools for Content Security Policy (CSP), CORS, and Web Messaging APIs.
- **Architected Epic's clinical decision support (CDS) web services** leveraging HL7 FHIR, CDS Hooks, and CDA standards. Redesigned configuration models to align with clinical mental models and implemented efficacy feedback loops for providers.
- **Recognized as a Subject Matter Expert** in Web App Security and RESTful integration at Epic, providing technical oversight for cross-application security standards and third-party vendor onboarding.

Research Assistant, Condor Distributed Computing

University of Wisconsin - Madison, WI *SEPT. 2008 - MAY 2010*

Virtualized Condor jobs with VMware achieving deterministic runtime environments.

Software Engineer, Enterprise Routing

Cisco Systems - Bangalore, India *JUNE 2006 - JULY 2008*

Platform engineering and configuration tooling for Cisco's enterprise class routers.

INDUSTRY & OUTREACH

- **HL7 FHIR Connectathons:** Developed prototypes of HL7 specifications in collaboration with participants from other vendors to improve interoperability.
- **HL7 Da Vinci Burden Reduction:** Represented Epic on public calls steering development of standards to reduce administrative burden in Prior Authorization.
- **HL7 CDS Hooks and SMART on FHIR:** Contributed to HL7 CDS Hooks and HL7 SMART Web Messaging specifications to enhance workflow integration and security.
- **Epic App Orchard Conferences:** Presented on CDS Hooks and CDS web service integration capabilities to educate and onboard third-party vendors.
- **Epic Recruitment:** Represented Epic at university career fairs and conducted candidate interviews to attract top talent to Epic.

EDUCATION

Master of Science, Computer Science

University of Wisconsin - Madison *2008-2010*

GPA: 3.94 / 4.0 | Focus: Systems, Networking, Distributed Computing.

Bachelor of Technology, Computer Science and Engineering

National Institute of Technology - Calicut, India *2002-2006*

CGPA: 9.60 / 10.0 | Department Topper and Gold Medalist.

This Resume!

2026 Q1

Think different! Resume developed as print-media web application with AI assistance.

github.com/mattvarghese/my-resume

SKILLS

Architecture & Infrastructure

- SDLC • Distributed Systems
- Microservices • Azure • CI/CD
- Docker • Virtualization • Hadoop
- Maintainable Architecture

Engineering Stack

- TypeScript • React • Web API
- Node • .NET/C# • Git & SVN
- SQL & NoSQL • Linux & Windows

Interoperability & Standards

- HL7 FHIR • SMART on FHIR
- CDS Hooks • CRD/DTR/PAS
- ANSI X12 278 • OAuth2
- JSON/JWT/JWK

Specialized Expertise

- LLM API Integration
- Agentic Workflows
- Network Architecture & Routing
- Encryption & Auth standards

EPIC CERTIFICATIONS

Web Tech Camp ⁝ Server Tech Camp ⁝ Clarity Data Model ⁝ EpicCare Inpatient Suite ⁝ Order Transmittal ⁝ Referrals and Authorizations ⁝ Tapestry Health Plan Suite

OTHER

- Novice (H2) rated hang-glider pilot
- Solo endorsed VFR student pilot
- ASA 101,103/104,105 rated sailor
- Learned Elementary Russian as adult
- Avid reader, ice-skating, volleyball, XC-skiing, bicycling